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Real-Space Deterministic Water Surface Dynamics via Temporal Recording of Local Direction Alternating Force

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Abstract

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Title: Real-Space Deterministic Water Surface Dynamics via Temporal Recording of Local Direction Alternating Force

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Abstract: I propose a deterministic framework for describing water surface dynamics based on the temporal recording of localized force–mass interactions. The surface response is interpreted as a time-ordered execution of local mechanical histories governed by causal accessibility. Numerical simulations constructed from this formulation reproduce structured surface patterns under geometric confinement and show close correspondence with experimentally observed water surface behavior. These results suggest that water surface patterns may be naturally interpreted as geometric records of force transmission over time.

Main Text: The dynamic behavior of water has historically been successfully described and interpreted through the lens of wave mechanics, governed by the three-dimensional Laplace equation for velocity potential. Comprehensive theoretical frameworks, encompassing both linear and nonlinear analyses, have provided robust models for these complex phenomena. Furthermore, recent advancements have demonstrated that these governing equations can be effectively leveraged to implement precise control technologies, such as through the use of artificial structures and metamaterials [1, 2]. Despite the water surface is a collective flow of discrete H2O molecules, the language describing water surface dynamics through forces acting on mass over time is limited.

Here, I explore whether a different point of view may be taken in describing the behavior of a water surface. Rather than beginning with a wave-based description or a predefined spatial operator, I consider the surface as a collection of local material elements whose motions evolve in time and interact through the accumulation of mechanical influence. From this perspective, the patterns observed on a water surface are not treated as entities that propagate through space, but as time-dependent records emerging from distributed local responses. In this view, the familiar structured patterns of surface motion arise naturally from the temporal organization of these local interactions, without presupposing a specific wave formalism or governing spatial equation.

In this work, I formulate a deterministic framework in which the evolution of surface patterns is governed by local vector-based rules operating under Newtonian dynamics and Hooke's law. The model is constructed from three fundamental principles: causal force propagation, geometric recording of force history at the front, and local regeneration at interaction boundaries. Together, these principles define a field that records time-dependent mechanical influence, allowing complex surface behavior to emerge without invoking a wave equation or predefined spatial operator. The following sections introduce the structure of this framework and its governing equation, before examining its implications through numerical simulations.

Conceptual origin of deterministic surface dynamics

In this system, the observed water surface response is decomposed into three sequential mechanisms: the generation of a source force, the temporal history of force transfer governed by a causal horizon, and the regeneration of the source through deterministic energy conversion (Figs. 1). The collective surface behavior of water molecules emerges from the time-ordered execution of these local mechanisms. Consequently, the apparent propagation of water surface patterns during propagation reflects a recorded sequence of local force states rather than a transported physical entity.

Source force refers to a localized mechanical input applied to the water surface, acting on a finite mass of fluid and introducing a well-defined force direction. Upon application, the surface element undergoes a mechanical displacement, and a restoring response arises due to elastic and gravitational effects inherent to the fluid interface. This initial force application sets the characteristic scale and orientation of the subsequent surface motion, while the resulting behavior evolves through the deterministic response of the local fluid mass. In this context, the applied force establishes an upward deformation of the free surface, which serves as the geometric origin for subsequent surface propagation.

Force transfer describes how the mechanical effect of a source force applied at the water surface becomes manifested at neighboring surface regions over time. Following the initial force application, adjacent portions of the water surface experience delayed mechanical responses as the disturbance reaches finite masses of fluid. This transfer process is constrained by a finite propagation speed, which determines the time at which each surface element begins to respond to

the applied force. During this process, the transferred effect does not involve the continuous transport of a force vector across the surface. Instead, it reflects a sequence of local energy conversions, in which directionally encoded potential energy is repeatedly transformed into kinetic response at neighboring surface regions. Once this condition is established, each surface region evolves through local force–mass interactions governed by the same mechanical relations as the source region, without requiring sustained external forcing.

Force regeneration indicates the re-emergence of surface motion following mechanical interaction with boundaries or geometric constraints and does not imply the creation of a new independent source, but rather denotes the continued execution of the original force history under altered geometric conditions. When a propagating surface deformation encounters a barrier, the horizontal motion is arrested, and the surface geometry is locally reconfigured. The impact force, together with gravity, converts the incoming kinetic response into a downward-directed potential energy stored within the local fluid mass.

This stored potential energy subsequently relaxes through the same restoring mechanisms that govern the initial surface response, giving rise to a renewed upward surface motion. Importantly, this process does not involve a reversal of the propagation direction, but rather a repetition of the same force–mass interaction under altered geometric conditions. As a result, surface propagation is regenerated through deterministic energy conversion, in which an upward deformation of the free surface is first re-established and subsequently enables further propagation, rather than through reflection or amplification.

A local deterministic causal framework of surface motion

Before introducing the governing equations, it is important to clarify the conceptual origin and structural intent of the present formulation. The mathematical framework employed here emerged through a gradual refinement of a previously proposed description of force-mediated surface propagation [3]. In that work, an eikonal-type formulation was used as a temporal registration device, encoding when a given spatial location becomes mechanically eligible to respond to a localized force input [4, 5]. The present study builds directly on that conceptual foundation, while extending it to a fully deterministic force–mass framework applicable to real water surfaces. Here, the eikonal front is retained not as a wave construct, but as a causal horizon that governs the temporal ordering of local mechanical responses. This allows the surface evolution to be described through a sequence of local force applications, transfers, and regeneration, all governed by classical mechanics. Accordingly, the full governing equation introduced below is constructed to reflect the three mechanisms outlined in the system description: (i) localized source force generation, (ii) temporally ordered force transfer defined by a finite propagation rule, and (iii) deterministic regeneration through energy conversion within the local fluid mass. The equation is presented first in its complete form, after which each term is introduced in relation to the corresponding physical mechanism. To describe the observed collective dynamics of the water surface, we construct a minimal deterministic model that directly reflects the three sequential mechanisms identified in the system: source force generation, time-recorded force transfer, and force regeneration. The governing equation is formulated to encode how a localized mechanical input becomes accessible across space through time and is locally executed by the fluid mass.

$$\Psi\left(r,t\right)=A\alpha\left(\tau\right)H\left(\phi\right)$$

Here, $\Psi(r, t)$ represents the recorded mechanical state of the water surface at position r and time t. Importantly, ψ does not denote a transported physical quantity, but the outcome of a locally executed force–mass interaction conditioned by the accessibility of prior force history.

Source Force Term A

The coefficient A represents the magnitude of a localized mechanical input applied to the water surface at position r_s. This force acts on a finite mass of fluid and introduces a well-defined initial direction to the surface displacement. The applied force establishes the characteristic scale and geometric orientation of the subsequent surface motion. After this initial input, no sustained external forcing is required; all subsequent dynamics arise from the deterministic mechanical response of the local fluid mass.

Local Mechanical Response $a(\tau)$

The function $a(\tau)$ describes the time evolution of the local surface response following force application:

$$a(\tau) = e^{-\gamma\tau}\cos(\omega\tau), \qquad \text{with } m \cdot \ddot{a} + c \cdot \dot{a} + k \cdot a = 0$$

This term corresponds to the solution of a damped harmonic response governed by the effective mass m, damping coefficient c, and restoring stiffness k of the water surface. Physically, it represents how a displaced surface element undergoes restoring motion due to gravitational and elastic effects. The response is entirely local and deterministic, and its temporal evolution is recorded rather than propagated.

Causal Horizon Operator $H(\phi)$

The term $H(\phi)$ defines the spatiotemporal accessibility of the recorded force history:

$$H(\phi) = \Theta\left(t - \frac{|r - rs|}{V}\right)$$

where V is the finite propagation speed governing the expansion of causal accessibility. This operator does not transport force or energy across space. Instead, it determines whether a given surface location can access the force history up to time τ. As a result, different spatial locations execute identical mechanical rules, but over different accessible time intervals. Within this framework, the apparent propagation of water surface patterns arises from the spatial expansion of temporally accessible force histories. Each surface element responds by executing the same local force–mass relation once the causal condition is satisfied. When the accessible force history reaches a geometrical boundary, the system does not undergo force reversal. Instead, the boundary initiates a new localized source force, restarting the same deterministic process under altered geometric constraints. This regeneration mechanism preserves the directionality of the force–mass interaction while allowing continued surface propagation.

Numerical realization of surface motion under local deterministic causality

Figs. S1(A) and Movies S1 present the numerical simulation of surface evolution generated solely by the governing equation, without imposing any boundary-induced regeneration or external constraints. A single localized source force is applied at the center of the domain, and the subsequent surface response is computed according to the deterministic force–mass dynamics and causal accessibility defined in the model. The upper panels show the two-dimensional surface record $\Psi\left(r,t\right)$ at increasing times. As time progresses, a radially expanding surface pattern emerges from the source location. This expansion arises from the sequential activation of local surface elements as their accessible force history becomes available. The spatial extent of the pattern increases monotonically with time, reflecting the finite speed at which causal accessibility expands across the surface. The lower panels display one-dimensional cross-sectional profiles taken through the center of the domain. These profiles reveal a structured alternation in the recorded surface state as a function of position, indicating that each surface element executes a time-dependent mechanical response once activated. Notably, the spatial repetition observed in the cross-sections is not preassigned but emerges from the combination of local restoring dynamics and the temporal ordering imposed by the causal horizon. A characteristic feature of the simulation is the asymmetric structure at the leading edge of the expanding surface record. The front exhibits a truncated or partially developed structure. This behavior reflects that surface elements near the advancing front access only a limited portion of the force history and simultaneously experience a spatially distributed reactive force arising from local mechanical recoil. As a result, the local response near the front remains geometrically and temporally incomplete compared to regions closer to the source.

Figs. S1(B) and Movies S2 examines how variations in the initial mechanical conditions influence the resulting surface record. Compared to Figure S1(A), the input force is increased, the effective mass is reduced, and finite damping is introduced. These changes localize the initially perturbed surface region while enhancing the directional magnitude of the response. Under these conditions, the leading surface record exhibits a pronounced directional amplitude aligned with the initial force orientation, whereas subsequent surface responses are strongly attenuated. This contrast indicates that the earliest activated surface elements retain a dominant portion of the force history, while later responses are increasingly suppressed due to geometric dispersion and damping. The cross-sectional profiles emphasize that the spatial distribution of recorded surface states is highly sensitive to the initial force–mass configuration. Small changes in the applied force and local inertia can produce qualitatively different surface evolution patterns, even under identical governing rules. This result highlights that surface propagation in the present framework is not universal, but depends explicitly on the mechanical conditions under which the source force is applied. Figure 2(A) and Movies S3 show the surface record produced by a single localized source force under finite damping, illustrating the basic source-driven local force–mass response introduced earlier. Figure 2(B) and Movies S4 display the surface records generated by two localized source forces. The resulting patterns exhibit spatial regions where local mechanical responses are enhanced, as well as regions where they are locally diminished. These features arise without the introduction of additional mathematical rules or coordination mechanisms and follow directly from the interaction of independently executed local force histories. Where accessible force records encode compatible directional tendencies, the local response is reinforced; where opposing directional tendencies coexist, the response is reduced. The observed structure reflects the directional nature of locally recorded force–mass interactions and their concurrent execution within overlapping regions of temporal accessibility. This behavior indicates that the mechanical

record retains directional information and that local force executions can coexist and interact in space without requiring global coordination.

Before introducing boundary interactions, I briefly specify the numerical conditions under which the confined-domain behavior is examined. To maintain consistency with the present formulation, boundary-induced regeneration is implemented as a single event at each geometric constraint. In addition, dissipation is set to zero so that the force history originating from the initial localized event remains continuously executable in time. Under these conditions, surface confinement arises solely from geometric limitation and the persistent execution of the original force history, rather than from repeated excitation or algorithmic boundary rules. Figs. 3 and Movies S5 present the surface records generated by the present formulation when the spatial domain is geometrically confined. Without introducing any additional constraints or organizing rules, the localized force–mass interactions produce stable, spatially ordered surface structures. The two-dimensional surface records exhibit repeating geometric patterns, while the corresponding one-dimensional cross-sections reveal regular alternation in the recorded mechanical state. These structures arise solely from the deterministic execution of local force histories over time. Each surface element responds according to the same local mechanical rule, while the spatial confinement limits the accessible force record and thereby shapes the resulting surface organization. The observed regularity does not result from imposed spatial coherence, but from the accumulation and interaction of directionally encoded force histories within a bounded domain. The emergence of such stable and ordered surface records indicates that the present formulation captures essential mechanical features governing water surface behavior. The results demonstrate that complex and repeatable surface organization can arise naturally from local force execution and temporal ordering alone, without invoking non-local coordination or additional descriptive constructs.

Figs. 4 compares surface records generated by the present formulation with experimentally observed water surface behavior under comparable geometric conditions. In panels (B) and (D), water droplets are introduced at the side and center of a confined container, respectively, and the resulting surface motion is recorded over time. Panels (A), (C) and Movies S6, 7 show the corresponding numerical surface records produced by a single localized source in geometrically similar domains, without introducing additional boundary rules, reflection conditions, or interaction terms beyond the governing formulation. Despite the absence of explicit boundary-specific or phenomenological corrections, the simulated surface records exhibit spatial and temporal patterns that closely resemble those observed in the experiments. In particular, the geometric organization of surface structures imposed by the container boundaries is reflected in both the experimental images and the recorded force patterns generated by the model. This correspondence indicates that the present formulation captures key mechanical features governing water surface behavior through local force execution and temporal ordering alone. The observed correspondence suggests that describing water surface motion as a time-recorded execution of local mechanical interactions provides a natural and effective framework for interpreting experimentally observed surface patterns. Across both experiment and simulation, the governing formulation is unchanged. The only distinction lies in the spatial placement of the initial localized event, from which all subsequent surface patterns emerge through time. Figs. S2 and Movies S8 show surface records generated by multiple localized source events introduced at randomized spatial locations and initiation times. No additional interaction rules, coordination mechanisms, or boundary-specific conditions are imposed beyond the governing formulation. Each source executes the same local force–mass rule, and all surface structures arise from the temporal overlap and spatial accessibility of their respective force histories. Despite the absence of imposed organization, coherent surface patterns emerge and evolve through time. This result demonstrates

that the present formulation supports the formation of complex surface structures through local deterministic interactions alone, even under heterogeneous and uncoordinated initial conditions. The observed surface organization reflects the robustness of the formulation, indicating that pattern formation arises from the execution of local force histories rather than from prescribed global order.

Conclusion and perspectives

This study demonstrates that a minimal, history-dependent force formulation is sufficient to reproduce complex surface patterns observed in confined water systems, without invoking conventional wave equations, interference rules, or material-specific boundary conditions. Across simulations and physical experiments, similar geometric surface structures emerge solely from the temporal evolution of force propagation and the spatial location of its origin, indicating that pattern formation is governed by accumulated force history rather than predefined wave mechanics. The agreement between simulated force records and experimentally observed water-surface motions suggests that the proposed formulation captures an essential aspect of how real fluids encode time and geometry into observable patterns. Taken together, these results support the view that surface dynamics can be interpreted as a geometric record of force transmission over time, providing a compact and alternative framework for describing pattern formation in fluid systems.

References and Notes

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Diversity, equity, ethics, and inclusion [optional]:

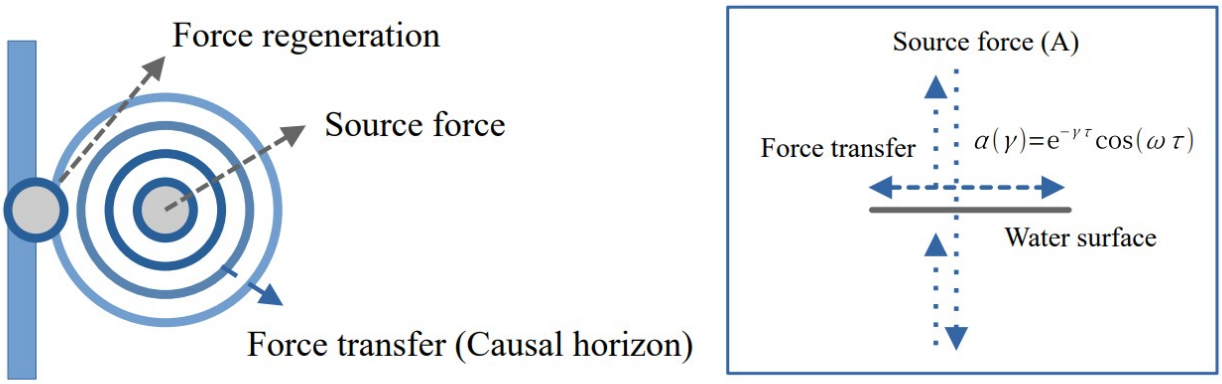
Competing interests: Author declare that they have no competing interests.

Data and materials availability: All data are available in the main text or the supplementary materials. The source code is available at: https://github.com/junhyunglee73/real-space-water-surface-dynamics.git

Supplementary Materials

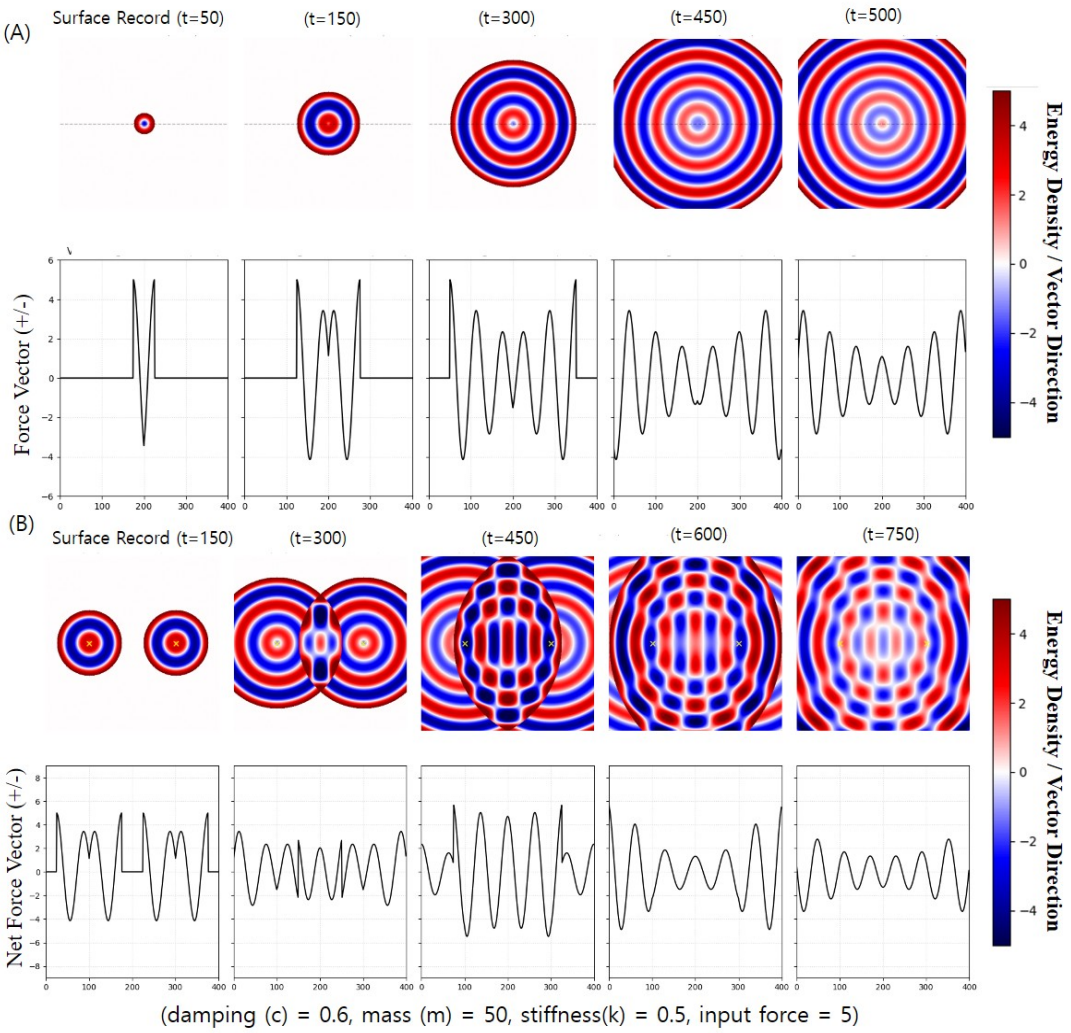
Experiment

Figs. S1, S2

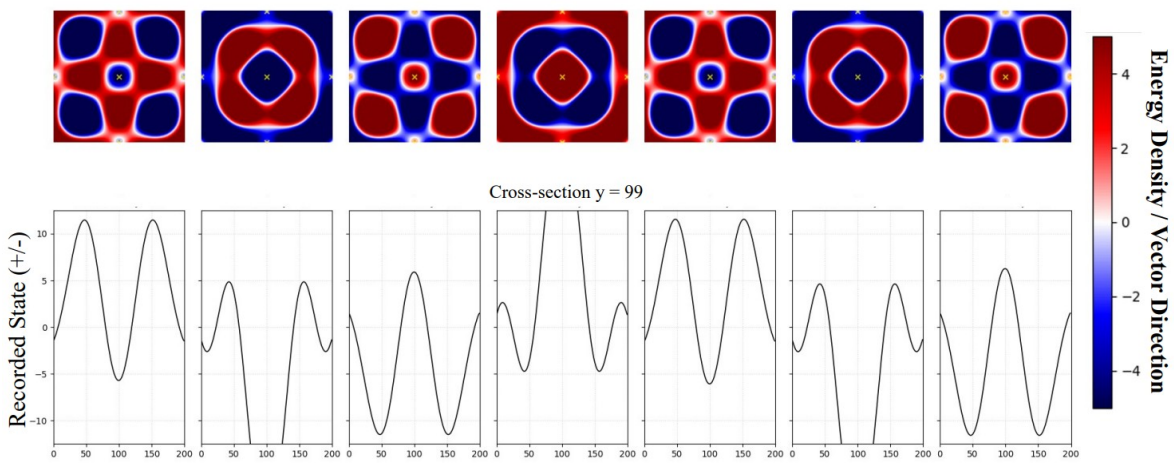


$$\Psi(r,t)=A\cdot a(\gamma)\cdot H(\phi)$$

Figs. 1. Conceptual schematic of deterministic surface dynamics. The surface response is decomposed into three sequential mechanisms: (i) generation of a localized source force, (ii) force transfer constrained by a causal horizon that defines information accessibility in time and space, and (iii) force regeneration through deterministic energy conversion and directional reconfiguration.

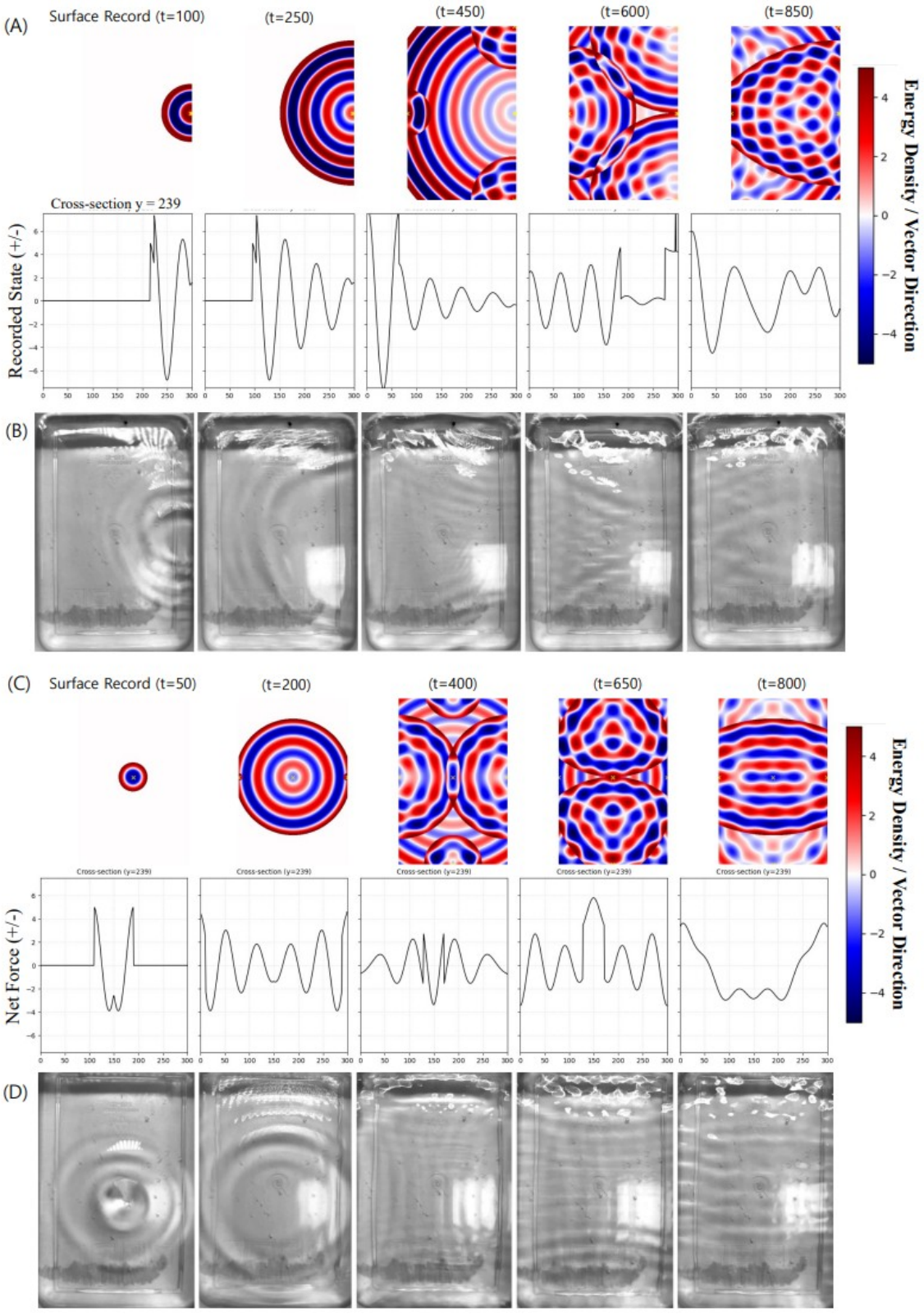


Figs. 2. Deterministic surface record generated by (A) single and (B) dual source forces



(200 x 200, damping (c) = 0.0, mass (m) = 100, stiffness(k) = 0.5, input force = 5)

Figs. 3. Source-induced periodic surface records under geometric confinement



Figs. 4. Comparison between simulated surface records and experimental water surface pattern (A,C) Simulated surface records generated by the present formulation in geometrically confined domains with localized sources placed at the side and center, respectively. (B,D) Corresponding experimental water surface patterns obtained by introducing droplets at analogous positions in a container of similar geometric proportions.

Supplementary Materials for

Real-Space Deterministic Water Surface Dynamics via Temporal Recording of Local Direction Alternating Force

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The PDF file includes:

Experiment
Figs. S1, S2

Experiment

Water surface patterns formation

Water was poured into a plastic container (19 × 12 cm), and surface patterns were generated by introducing droplets at locations corresponding to those used in the simulations. The resulting surface motion was recorded using the camera of an iPhone 14.

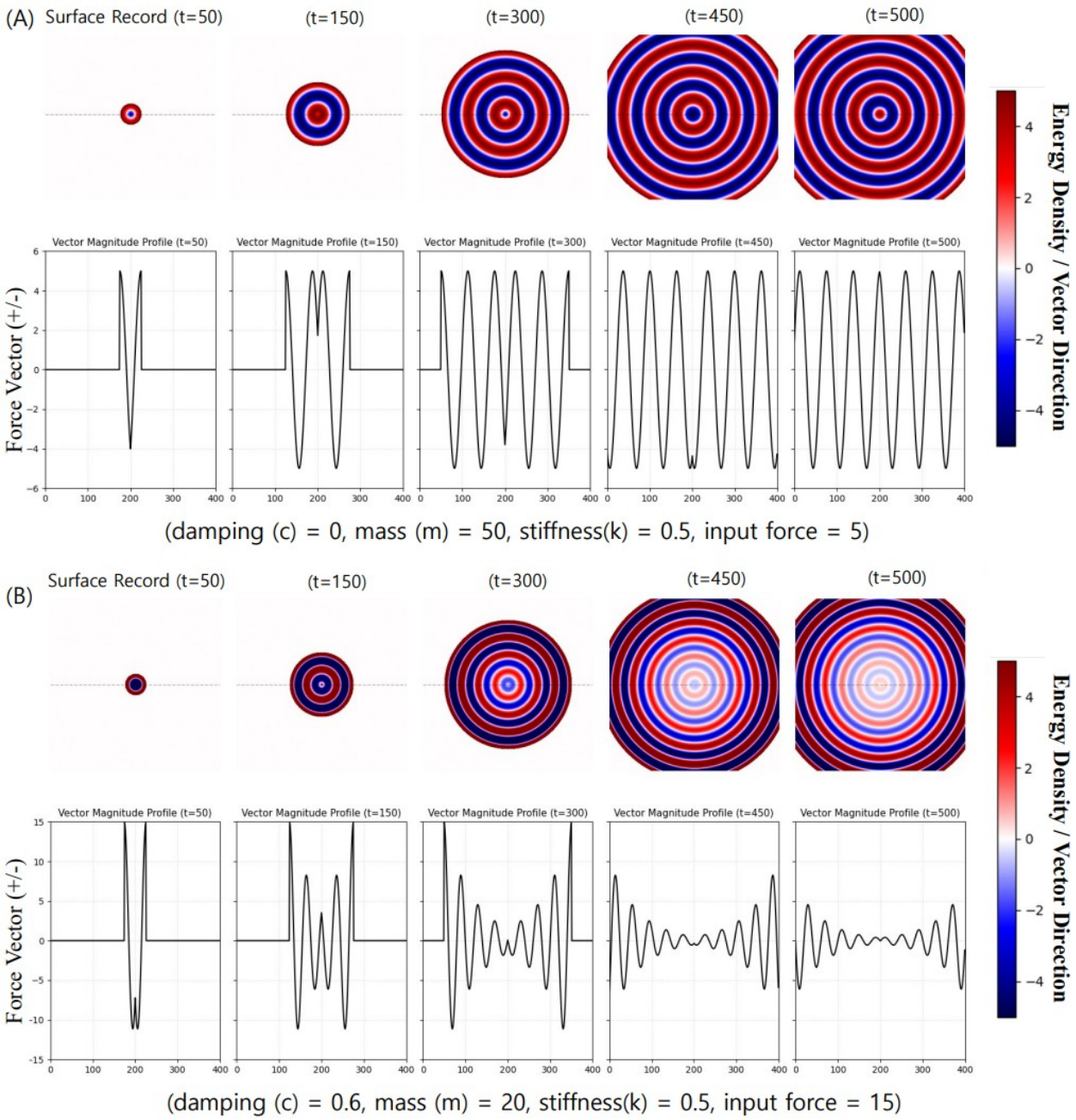


Fig. S1. Emergent surface patterns generated by a single causal source in the absence of boundary regeneration. (A) Time-resolved snapshots of the recorded surface state produced by a localized source force governed solely by the causal record kernel. No boundary-induced source regeneration or additional spatial operators are applied. B) Increasing the source force and reducing the effective mass modifies the directional strength of the initial surface response, demonstrating that the propagation pattern is directly shaped by the input force parameters rather than imposed spatial constraints.

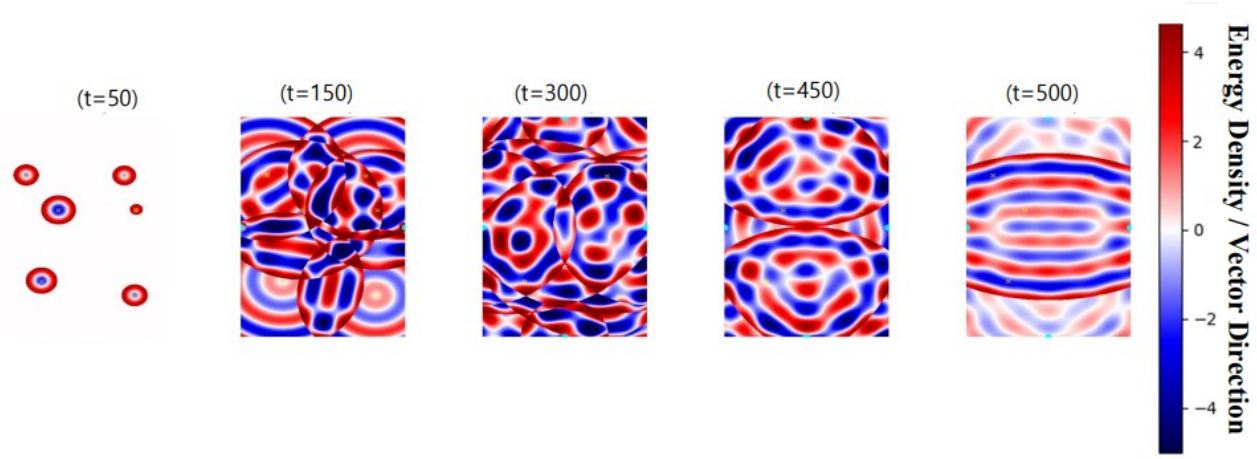


Fig. S2. Surface records generated by multiple randomized localized events. All surface structures arise from the temporal overlap and spatial accessibility of force histories generated by randomly placed and timed localized events.